

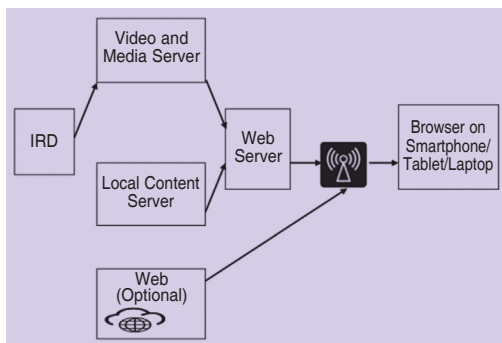


Fig. 1: Block diagram for converged gateway node network (Source: IIT Kanpur June 2022 white paper)



Broadcast radio heads and USB dongle: key D2M equipment parts from Saankhya Labs (Source: <https://saankhyalabs.com>)

Additionally, the project offers strategic capabilities to deliver emergency alerts and disaster management content to millions of smartphones and other mobile devices in a targeted manner.

The Departments of Information and Broadcasting, Science and Technology, Telecom, and Electronics and Information Technology are collaborating with IIT Kanpur to implement the D2M project.

IIT Kanpur and Saankhya Labs are spearheading the project using software-defined radio technology, the NextGen P3 chip, broadcast radio heads, reference smartphones, USB dongles, and Wi-Fi gateways developed by Saankhya Labs.

Nineteen pilots are currently being conducted in cities such as Bengaluru, New Delhi, and NOIDA, leveraging Prasar Bharti infrastructure to test the technology. Full deployment is expected by 2025, upon successful completion of the trials.

The technology deployed

The D2M technology used in the project is based on the NextGen broadcast architecture and aligns with the standard TEC 57040:2023 'generic requirement (GR) on converged gateway node for delivering broadcast content to portable devices through wireless LAN.' This standard was released by the Telecommunication Engineering Centre (TEC).

The converged gateway node network receives terrestrial or satellite signals, which are demodulated, decoded, and selectively streamed over WLAN/Wi-Fi. The audio and video content is made available to mobile devices, allowing end users to access it without relying on any specialised software, plugin, or mobile data. The system layout is shown in Fig. 1.

The converged gateway node is also adaptable for use in other projects, such as BharatNet and PM-WANI. Prime Minister Wi-Fi Access Network Interface (PM WANI) scheme has been launched by the DoT.

Integrated receiver decoder (IRD).

The IRD uses a dish antenna, located on the ground at the gateway, to receive downlinked signals from the satellite. A smaller antenna is used to receive terrestrial signals. A device at the dish antenna amplifies the received signals and converts them into a suitable band, which is fed to demodulation devices for conversion into baseband signals, such as multiple digital TV or radio channels. These are then streamed to the video and media

server. Each demodulation device is tuned to a specific frequency to receive a set number of channels. Multiple devices are employed to accommodate all satellite and terrestrial channels.

Video and media server. The streams generated are not playable by industry-standard browsers. The video and media server transcodes the streams, making them HTML-compatible with browsers. It also ingests streams, breaking them into files hosted over HTML, enabling audio and video content to be accessed without requiring any specialised software or plugin. Third-party plugins are not only inconvenient but also pose security risks. This solution enhances user convenience and security by eliminating the need for plugins.

Self-learning web server. The self-learning web server serves two primary functions. First, acting as an HTML or web server to provide audio and video content to end users. Second, learning from user viewership statistics to generate a list of the most-watched channels. Initially, channels are selectively streamed for a short period. A self-learning algorithm then determines the most desired or watched channels at a given location. Once the algorithm converges, the optimised list is used for streaming. The self-learning web server also reduces the number of demodulating devices required in the IRD.

End-user browser. Users connect to Wi-Fi, access the web portal via a provided link in their browser, browse the channel list, and play content by selecting the channel name.

BharatNet Project and PM WANI Scheme

BharatNet. BharatNet, one of the biggest rural telecom projects in the world, is being implemented by the government of India to cover *Gram Panchayats* (approximately 250,000) to provide them non-discriminatory broadband connectivity. The project can use the converged gateway node to enable the access providers like mobile operators, internet service providers, and cable TV operators to launch services like e-health, e-education and e-governance in rural and remote regions.

PM WANI. To enervise broadband internet proliferation, Prime Minister Wi-Fi Access Network Interface (PM WANI) scheme has been launched by the Department of Telecommunications. The scheme provides public Wi-Fi hotspots throughout the country to drive up the connectivity alternatives and enhance digital access. The converged gateway node technology can be deployed gainfully in the scheme to make broadcast channels available in regions where internet is not available.